Aaditya Pratap Singh Rana

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github.com/respondworld00

Professional Summary

Final-year Computer Science student with strong fundamentals in data structures, algorithms, and object-oriented programming. Experienced in building scalable applications and machine learning models with a focus on clean and maintainable solutions.

Technical Skills

Languages: Java, JavaScript, Python, Dart, SQL, C++
Mobile & Frontend: Flutter, Dart, Cross-platform Development, Material Design
Cloud & Tools (Basics): AWS (EC2, S3), Git, Docker (Basics), Linux (Basics)
Machine Learning: Regression Models, Classification, Clustering, CNNs, Feature Engineering
Core CS: Data Structures & Algorithms, OOP, Database Design, System Design

Experience

Machine Learning Intern February 2025 – March 2025
Prodigy InfoTech

- Developed regression models to predict housing prices with 85% accuracy through systematic feature engineering and hyperparameter tuning, improving model performance by 12% over baseline.
- Implemented K-Means clustering to segment customer data into 5 distinct groups, reducing processing time by 30%.
- Trained CNNs for image classification achieving 92% validation accuracy using custom architecture and data augmentation techniques.
- Structured and cleaned datasets with 50,000+ records, reducing preprocessing errors by 40%.

Projects

Smart Kitchen Management System

- Built full-stack application supporting 1000+ concurrent users for inventory, recipes, and meal planning.
- Designed relational database schema with optimized indexing, reducing average query response time from 200ms to 45ms.
- Implemented JWT-based authentication and authorization middleware ensuring secure API access.
- Developed automated inventory tracking with real-time alerts, reducing food waste by 25%.

Cross-Platform Mobile Application (Flutter)

- Developed a cross-platform mobile application using Flutter and Dart for Android and iOS from a single codebase, reducing overall development time by 50%.
- Integrated RESTful APIs for real-time data synchronization using Provider pattern for efficient state management.
- Designed responsive UI components following Material Design principles across multiple screen sizes.
- Implemented SQLite-based local data persistence enabling offline functionality and reducing server load.

Education

Raj Kumar Goel Institute of Technology, Ghaziabad
Bachelor of Technology in Computer Science October 2022 – August 2026

Geeta Sanjay Memorial Public School
XII (Senior Secondary) 2022

Ch. Chhabil Dass Public School
X (Secondary) 2020